

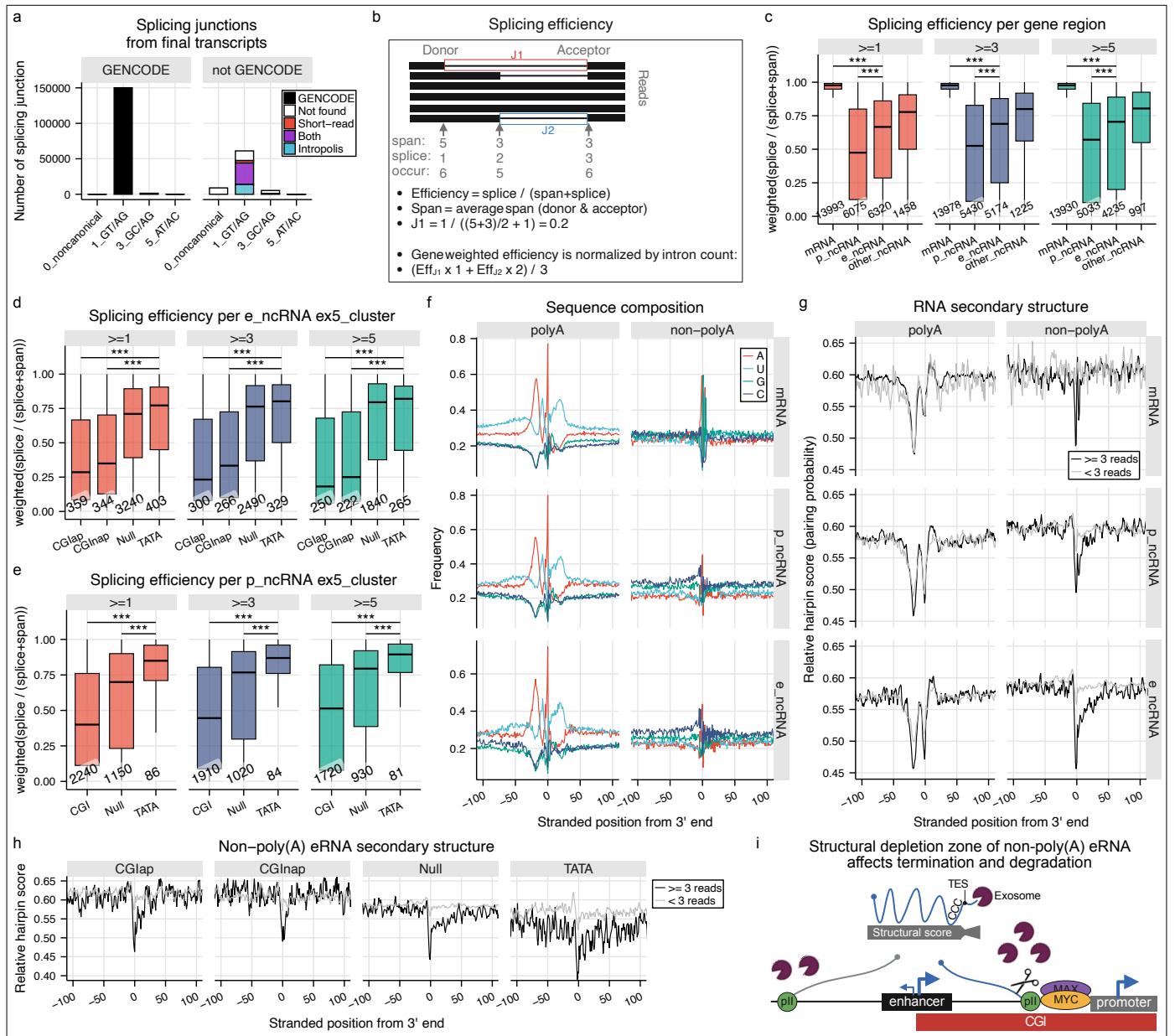


Figure 5 | Splicing efficiency and transcription end sites of non-poly(A) ncRNAs.

a, All splice junctions (SJ) identified in the final transcriptome. Most were within GENCODE (v39) annotation, while the majority of novel (non-GENCODE) SJs were canonical. **b**, Schematic illustrating the calculation of splicing efficiency per junction and the weighted splicing efficiency per gene. **c**, Splicing efficiency summarized at the gene level and normalized by intron count (see Methods and panel **b**). Gene models lacking splice junctions were excluded. SJs were filtered using three thresholds for junction support (≥ 1 , ≥ 3 , or ≥ 5 total reads spanning or splicing both donor and acceptor sites). **d**, Splicing efficiency of e_ncRNAs grouped by ex5_clusters and compared across regulatory elements (CGIap, CGIap, Null, TATA box). Only ex5_clusters assigned with one gene model were kept. **e**, Same analysis as in **d** for p_ncRNAs. **f**, Nucleotide composition surrounding recursive transcription end sites (TESs; ≥ 3 read support) of different RNA classes, stratified by poly(A) status. **g**, RNA secondary structure predictions from RNAfold using ± 200 nt sequences centered on the 3' ends of poly(A) and non-poly(A) eRNAs. Relative hairpin scores were summarized for ± 100 nt around the TES. Sequences were input to RNAfold without accounting for splicing. Analyses were conducted separately for recursive (≥ 3 reads) and non-recursive TESs. **h**, Same analysis as panel **g**, subgrouped by regulatory elements from non-poly(A) eRNAs. **i**, Schematic illustrating structural depletion-associated non-PAS termination of RNA derived from CGIap enhancer. Unless specified, statistical tests were two-sided Wilcoxon; $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant.